Play it again, Sam

Move on from MP3 to four of the most advanced audio formats of today and enjoy high fidelity and low file sizes

Long, long ago, in the decade of bad taste in the last century, a new audio format jumpstarted the audio revolution with the PC. Even today, almost two decades later, MP3 is almost the *de facto* standard for storing music with its promise of good CD Audio quality and small file sizes. But audio compression technology did not stop at MP3. Dramatic improvements in personal computing technology and rapid advances in the field of audio compression are moving towards that perfect audio compression within a small file size.

Audio compression 101

The quest is on for the compressed audio format that sounds every bit as good as the uncompressed audio file. An uncompressed CD Audio occupies roughly 10 MB of space for every minute, so the files have to be reasonably compressed to store them. To achieve such perfect compression, mathematical techniques are used to model the working and psycho-acoustic responses of the human ear.

Audio compression is of two types—lossless compression makes perfect copies of the original wave file when you uncompress the file, keeping the data intact, like a Zip file compression. Lossy file compression, on the other hand, distorts some of the data, causing the file to lose some information.

The human hearing range is most sensitive to frequencies between 2 and 4 KHz. Audio file compression is achieved by removing those inaudible sections of the pure audio file which lie above and below the human threshold of hearing. Typically, each codec uses complex and unique mathematical algorithms to shrink the size of a pure sound file, with a minimal loss of quality. Smaller files are, of course, advantageous for both storage and transferring between computers. While greater compression is possible, especially in the case of lossy codecs, it comes at the price of quality. Thus, an ideal balance between acceptable quality loss and small file size is needed.

Not all bits are created equal

Many formats have tried to improve upon the MP3 promise of high fidelity audio with smaller file sizes. Some of them are pretty much extinct with their developers having abandoned the standard, but a handful of plucky and battle-hardened formats have survived, each using a different way to cram in more music.

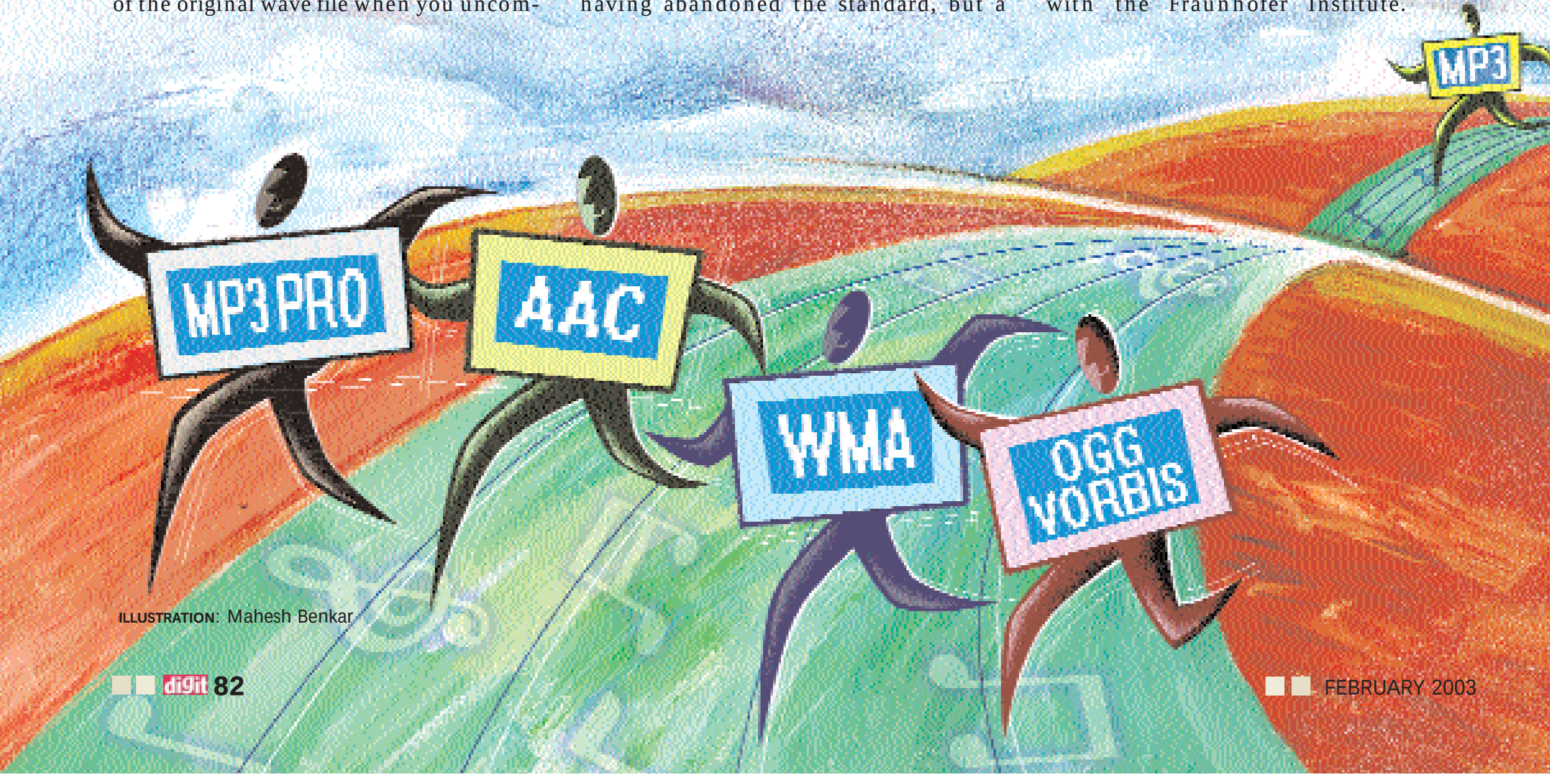
The two primary reasons that MP3 achieved cult status were the easy availability of MP3 files before the music industry drove the bootleg MP3 industry underground and the easy availability of quality MP3 playing software. For any format to take over from where MP3 left off, it will need to have these two factors and more. It will need to catch the interest of the developer community that democratised the MP3 revolution by creating shareware/free MP3 players, entire archives of MP3s online and peer to peer file swapping services.

Four formats are poised to take the baton from MP3 . Admittedly, they are not riding the popularity wave yet, but they could well emerge as dominant formats.

MP3PRO

Web site: *www.mp3prozone.com*

The proprietary MP3PRO standard was developed by Thomson Multimedia in 2001, and they share the patent rights with the Fraunhofer Institute.



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